

Relation of isoflavones and fiber intake in childhood to the timing of puberty.

BACKGROUND: It has been suggested that phytoestrogens and dietary fiber can affect puberty timing. **OBJECTIVE:** We examined whether intake of isoflavone and fiber in healthy white children before their pubertal growth spurt [age at take-off (ATO)] was associated with puberty timing. **DESIGN:** Multivariate regression analyses were performed in 227 DONALD (Dortmund Nutritional and Anthropometric Longitudinally Designed) Study participants with 3-d weighed dietary records and information on potential confounders at baseline (1 and 2 y before ATO). In a subsample (n = 111), urinary isoflavones were determined in 24-h urine samples by gas chromatography-mass spectrometry analysis. Puberty timing was examined by using ATO and chronologic ages at pubertal stage 2 for breast development (B2) or gonadal development, peak height velocity (PHV), and menarche or voice break. **RESULTS:** Girls whose diet was in the highest dietary isoflavone tertile experienced Tanner stage 2 for breast development ap 0.7 y later and reached PHV ap 0.6 y later than did girls whose diet was in the lowest isoflavone tertile [age (95% CI) at B2: 10.7 y (10.4, 10.9 y) compared with 10.0 y (9.7, 10.3 y), respectively; P for trend = 0.04; age at PHV: 11.9 y (11.6, 12.2 y) compared with 11.3 y (11.0, 11.6 y), respectively; P for trend = 0.04; adjusted for body mass index z score and fiber intake]. In boys, dietary isoflavones were not associated with pubertal markers. Urinary isoflavone and dietary fiber intakes were not associated with pubertal markers. **CONCLUSIONS:** Girls, but not boys, with higher prepubertal isoflavone intakes appear to enter puberty at a later age. Fiber intake in this sample of healthy white girls and boys was not relevant for puberty timing.